A Dynamic L-System-Based Architectural Maize Model for 3-D Radiative Transfer Simulation

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***Abstract*— We integrate the time series simulation capability of the maize model within an extended L-system (ELSYS) using the growth equations from a 4-D maize and a leaf breakpoint model. These models simulate maize growth from emergence to male anthesis, accounting for 3-D architecture during the vegetative season. We employ two methods to achieve time series simulation in ELSYS: directly use the growth equations in the 4-D maize and leaf breakpoint models, and name ELSYS coupling 4-D maize ($ELSYS_{4Dmaize}$). Alternatively, replace the stem radius-leaf order function with a stem radius-height function and employ linear interpolation to transform the leaf width-length ratio from a constant value to a function that varies with leaf order, thereby simulating a 4-D maize structure, and name the dynamic L-system-based architectural maize (DLAmaize) model. The DLAmaize model is applied to maize canopy reflectance simulations using the discrete anisotropic radiative transfer (DART) model and radiosity-graphics combined method (RGM), along with a comparison with the 1-D scattering by arbitrarily inclined leaves (SAIL) model. The simulated reflectance of maize canopy from $ELSYS_{4Dmaize}$ and DLAmaize differs significantly in the hotspot direction (the absolute value of the relative difference can be up to 67.4%). In addition, comparisons among RT models show that the DART model and RGM simulate close reflectances. The SAIL model yields significant differences (e.g., the absolute value of the relative difference can be up to 41.16% in the nadir direction) owing to its assumption of the homogeneous canopy. DLAmaize enhances remote sensing of dynamic 4-D vegetation canopies modeling and holds promise for remote sensing applications.**

***Index Terms*— Architecture, discrete anisotropic radiative transfer (DART) model, extended L-system (ELSYS), maize canopy, plant development, RT modeling.**

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I. Introduction

VEGETATION ecosystems play a crucial role in various biogeochemical cycles, such as water [1], carbon [2], and nitrogen [3]. Remote sensing (RS) data, with its high-frequency temporal and extensive spatial coverage [4], is widely used for studying vegetation functioning and biophysical properties [5], [6], [7]. The increasing demand for more accurate vegetation information from RS observations, potentially with different sun-sensor geometries, highlights the need for precise simulations of RS observations with bidirectional reflectance factor (BRF) models [8], [9], [10].

In contemporary studies, three primary categories of BRF models are commonly employed, contingent upon the chosen simulation approach. These categories encompass empirical/semiempirical models [11], geometric optical (GO) models [12], and radiative transfer models (RTMs) [13], [14]. Empirical/semiempirical models [11] and GO models [12], whereas offering computational efficiency, often entail compromises in accuracy owing to significant approximations within their light propagation algorithms and in the representation of landscape covers, such as maize plants and trees. In stark contrast, 3-D RTMs [13], [14], exemplified by the discrete anisotropic RT (DART) model [15], [16] and the radiosity-graphics combined method (RGM) [14], can deliver highly precise simulations of RS observations, provided they comprehensively account for the primary physical mechanisms contributing to RS observations. For instance, the DART model uses the RT equation to simulate observed radiance as a function of absorption, scattering, and thermal emission [17], [18], [19]. However, these models are accurate only if they use precise descriptions of landscapes [20]. Here, we consider vegetation canopies.

Accurate models of vegetation canopies depend on modeling techniques, typically based on direct and indirect methods [21]. Direct methods, like the 3-D digitizer [22] or photogrammetric techniques [23], provide precise architectural descriptions of the canopy but are labor-intensive and inefficient. Conversely, indirect methods, such as point quadrates [24], hemispherical photographs [25], or laser techniques [26],

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offer more global information on canopy structures through the inversion of basic gap fraction models. However, these models rarely address complex canopy architecture without a priori knowledge of some structural features [27].

A canopy structure model results from dynamic plant development at a specific phase, with development patterns that are generally consistent [28]. This consistency can be formalized using dynamic modeling [29]. Four-dimensional models of canopy geometry have been created by incorporating botanical knowledge to describe plant architecture and fractals and field measurements of geometrical parameters to describe the size and orientation of organs varying with time [30]. Current dynamic models are empirical or include some understanding of the botanical, phenological, and physiological principles that regulate plant growth and development [31]. For example, various plant simulation platforms based on Lindenmayer models (L-systems) [32], such as the growth-grammar-related interactive modeling platform (GroIMP), can construct the architecture of numerous plant species [33]. OpenAlea [34], [35] serves as a sophisticated platform for simulating the intricate 4-D canopy architectural development of both architectural models of development based on L-systems (ADEL)-Maize [36], [37] and ADEL-Wheat [38], [39]. This simulation capability is contingent upon various abiotic environmental factors. In addition, the botany and modeling of plant architecture and vegetation (AMAP) model [40], [41] of plant architecture utilizes botanical principles. However, the application of these models still lies more in botany.

With the continuous advancements in 3-D light detection and ranging (LiDAR) scanning and associated technologies [42], plant geometric structural data acquisitions are increasingly efficient and convenient. The integration of a 4-D maize structure model with point cloud data acquired through LiDAR presents a promising avenue. For instance, leveraging artificial neural networks generated by ADEL-Wheat [38] in conjunction with a LiDAR simulator allows for the estimation of green area index (GAI) based on ground-based LiDAR observations over wheat [42]. GAI is defined as the total one-sided area of green vegetation elements per unit of ground horizontal surface [42]. While studies [43], [44] successfully couple crop growth models with RS, there is currently limited research on coupling 4-D plant growth models. Consequently, plant simulation models hold significant potential in the realm of RS.

As one of the plant simulation platforms based on the L-system, the extended L-system (ELSYS) [45] can create static plant models (e.g., maize [46] and arid area plants [47]) for 3-D RS scene construction. Maize, a typical row crop [48], is among the most crucial staples and has witnessed a substantial surge in consumption over the past few decades [49]. However, this model only simulates plants in instantaneous morphology [10], [47] because it only uses geometrical and morphological parameters. In contrast, using growth governed by cumulative degree days (T_d), a 4-D canopy model [50] can provide a simplified maize architecture model (i.e., primary geometric shapes such as a quadrangular pyramid for the stem and an isosceles triangle for each leaf). Despite being crude, this maize representation improves the realism of maize canopy models compared with the turbid medium representation commonly used by RTMs to simulate maize canopy reflectance [51]. In addition, a maize leaf break model [27] can simulate the breaking effect of the maize leaves. Recent studies [52], [53] show the importance of canopy structure in signal simulation. Therefore, the growth equations of the 4-D maize and leaf break models can be instrumental in offering the capabilities of the ELSYS model for accurately simulating time series dynamics in maize canopies.

This study aims to enhance the time series simulation capabilities of the maize model within ELSYS [45] by encompassing architectural attributes throughout the entire vegetative growth period, specifically from emergence to male anthesis. We employ two distinct approaches to enable the capability for temporal simulation in the maize model within ELSYS. The first approach directly utilizes the growth equations from the 4-D maize [50] and leaf break [27] models. The second approach entails transforming the constants within the growth equations of the 4-D maize model [50] into dynamic functions. This transformation encompasses establishing the stem radius as a function of stem height and the leaf width-length ratio as a function of leaf order, leading to the development of the dynamic L-system-based architectural maize (DLAmaize) model. This model facilitates the generation of 4-D representations of maize plants, catering to the requirements of 3-D RTMs. The maize structural data derived from this model is subsequently imported into the DART model and RGM to simulate BRFs and for cross-validation. Furthermore, a comparative analysis of the simulated BRFs between the scattering by arbitrarily inclined leaves (SAIL) [54] model and the RGM [46], [47] is undertaken to underscore the imperative nature of BRF simulations using 3-D RTMs.

II. Datasets

We employed two maize geometry datasets obtained from two campaigns conducted in Changchun City, Jilin Province, China. The first dataset, termed the normal growth group, was collected during a campaign focused on the natural growth of maize [55]. The second dataset, termed the stress-impacted group, was collected during a campaign aimed at investigating the effects of extreme conditions on maize growth.

A. Normal Growth Group

The dataset for the normal growth group was sourced from the study [55]. The study area encompassed the Changchun Jingyuetan RS experiment station of the Chinese Academy of Sciences, Beijing, China. Data were collected from 11 sites distributed near Kaoshan town (four sites), Taipingshan town and Shuangchengpu town (four sites), Longwang town and Chenjiadian town (two sites), and Donglalatun town (one site) (Fig. 1).

Each of the 11 sites, covering approximately 5 ha, was planted with annual maize on day of year (DOY) 125 and harvested around DOY 288, in 2021. Maize parameters were

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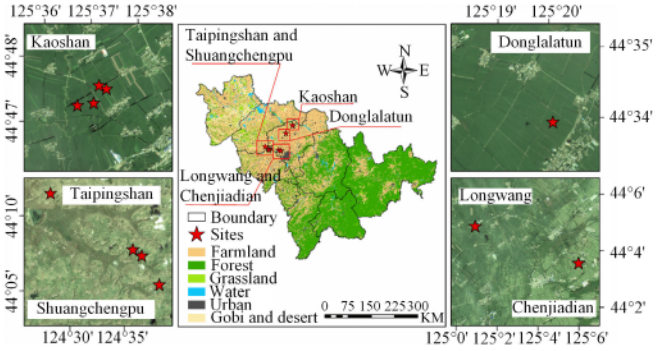


Fig. 1. Spatial distribution of the sampling points within the normal growth dataset, derived from the study [55], is made available under the terms of an open-access Creative Commons CC BY license.

TABLE I
CONTROL PARAMETERS OF THE STRESS-IMPACTED MAIZE EXPERIMENTS

Experiment	Parameter	Value
Fertiliser	Nitrogen	0 to 440, step 110 kg/ha
	Phosphorus	0 to 200, step 50 kg/ha
	Potassium	0 to 200, step 50 kg/ha
Waterlogging	Water control	3, 5, 7, 9 days
	Water deposit	1, 3, 5, 7, 9 days
Drought	ESD	S&J: 50%; F&M: 70% WFC
	MSD	S&M: 70%; J&F: 50% WFC
	LSD	S&J: 70%; F&M: 50% WFC
	CLD	S&J&F&M: 60% WFC
	CMD	S&J&F&M: 50% WFC
	CSD	S&J&F&M: 40% WFC
	NTR	S&J&F&M: 70% WFC

ESD: early-stage drought, MSD: middle-stage drought, LSD: late-stage drought, CLD: continuous light drought, CMD: continuous moderate drought, CSD: continuous severe drought, NTR: normal treatment reference. S: seedling stage, J: jointing stage, F: filling stage, M: maturity stage. WFC: water field capacity.

obtained by collecting field samples. Geometric parameters, including maize height and stem radius data, were measured on specific DOYs in Nong'an and Changling counties. The data collection occurred on various days throughout the year at distinct locations, specifically on DOYs 173, 198, 230, 242, 254, 266, 273, and 278 in the Nong'an county, and on DOYs 163, 175, 194, 218, 233, 247, and 266 in the Changling county. At each site, three plants were concurrently assessed, and the results were averaged to minimize measurement errors and accurately represent the regional maize growth states.

B. Stress-Impacted Group

The stress-impacted group investigated the impacts of fertilizers, waterlogging, and drought on maize growth. The maize, variety xianyu335, was sown in May 2019. It usually has a 127-day growing period, a plant height of 3.2 m, an ear height of 1.1 m, and approximately 19 leaves per plant. Measurements were taken throughout the maize growth phases. The fertilizer experiment was conducted outdoors, whereas the waterlogging and drought experiments took place in a greenhouse to regulate water content. Detailed parameters are in Table I.

The fertilizer experiment investigated the impacts of nitrogen, phosphorus, and potassium on maize growth. Nitrogen fertilizer, urea (N, kg/ha), ranged from 0 to 440 kg/ha in increments of 110 kg/ha; phosphorus fertilizer, calcium superphosphate (P_2O_5, kg/ha), ranged from 0 to 200 kg/ha in increments of 50 kg/ha; and potassium fertilizer, potassium sulfate (K_2O, kg/ha), ranged from 0 to 200 kg/ha in increments of 50 kg/ha. The test soil was sandy loam. Geometric canopy parameters were measured during the maize jointing, silking, filling, milk, and maturity stages. This fertilizer experiment is detailed in the study [56].

The waterlogging experiment investigated the impact of excessive water on maize growth. Two groups (water control and water deposit) were conducted: The water control group controlled the relative water content of the soil between 90% and 100%, with intervals of three, five, seven, and nine days. The water deposit group controlled watering deposits water above 0.03 m of the soil surface at intervals of one, three, five, seven, and nine days. The test soil type was black soil. Canopy geometry parameters were measured on the second day after stress cessation and 15 days after recovery. This waterlogging experiment is detailed in the study [57].

The drought experiment investigated the impact of water scarcity on maize growth. The water control treatment began after maintaining optimal soil moisture content during the emergence period (i.e., 65% of the maximum field water holding capacity), which ensured normal emergence for 25 days. The water control period was divided into seedling, jointing, filling, and maturity. Six continuous drought treatments were designed: early-stage drought, middle-stage drought, late-stage drought, continuous light drought, continuous moderate drought, continuous severe drought, and one normal treatment reference group. The test soil type was black soil. Canopy geometric parameters were collected at the end of each phase. This drought experiment is detailed in the study [58].

III. METHODOLOGY

A. Theory Background of the ELSYS and 4-D Maize Models

ELSYS [45] is a 3-D modeling platform with an L-system compiler. The input parameters of the static maize model based on ELSYS consist of geometric factors, such as stem height (H), bottom (R_b) and top (R_t) stem radius, leaf basal width (W_N) and leaf length (L_N), leaf attachment point height (H_N), leaf breakpoint position (P_{break}), and leaf inclination angle (θ_N) and leaf azimuth angle (φ_N), with N being the leaf order. As the ELSYS's output file format is only compatible with RGM [46] or thermal RGM (TRGM) [59], we designed a program to convert it to the standard 3-D wavefront format.obj commonly used by RTMs like the DART model. However, as ELSYS can only simulate maize based on maize geometric parameters, it can be coupled with the 4-D maize model [50] that provides maize growth equations to simulate the time series of the growth stage.

The 4-D maize model [50] provides a simple geometric representation of plant structure using a few canopy variables: maximum leaf area per plant ($S_{\max}$), maximum number of leaves per plant ($N_{\max}$), plant density (D), distance between rows (d_{rows}), leaf orientation plasticity, maximum stem height ($H_{\max}$), and plant growth phase driven by T_d. It can simulate

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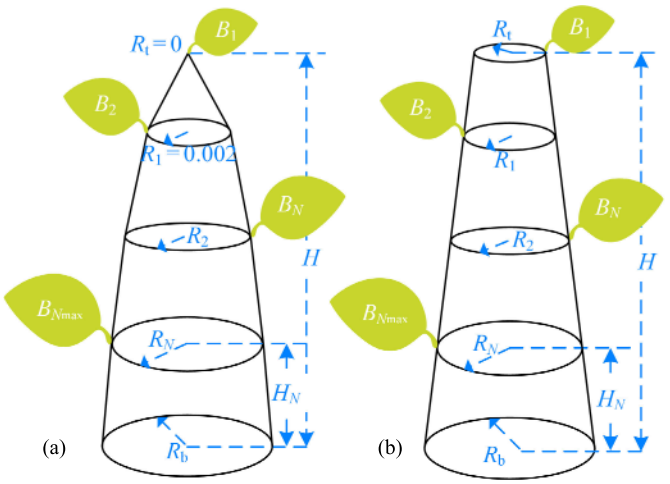


Fig. 2. Geometric representation of the relationship between maize stem radius (R_N) and leaf attachment point height (H_N) in the (a) 4-D maize and (b) ELSYS models. In the 4-D maize model, R_N at leaf (B_N) is linearly interpolated using the bottom stem radius (R_b) and the stem radius (R_1), combined with leaf order (N) and the maximum leaf order per plant ($N_{\max}$). In the ELSYS model, R_N is linearly interpolated using the R_b and stem top radius (R_t), combined with H_N and the stem height (H).

scenarios without requiring in-depth knowledge of canopy characteristics. Growth over time is regulated by N, which emerges as defined in the literature for maize canopies [60]. Two factors determine N: T_d since planting and phyllochron (T_d between the appearance of two consecutive leaves). Subsequently, the leaf phase can be converted into a calendar date. This 4-D maize model is detailed in the literature [50]. In addition, a leaf break model [27] is also coupled into the DLAmaize model. Appendix A presents the necessary equations from these two models that were used in this study. For example, stem radius varies with N (Fig. 2). By assigning the stem radius (R_1) at leaf $N = 1$ and R_b, the radius at other leaf orders can be calculated using linear interpolation (A.9). In addition, the ratio (P_b) of stem bottom radius (R_b) and height (H) is set at 0.008 for the entire growth stage. In contrast, in the 3-D maize model within ELSYS, the stem radius (R_N) is interpolated with the leaf attachment point height of leaf N (H_N) and the stem height (H) using the bottom (R_b) and top (R_t) stem radius, rather than N

$$R_N = \left(\frac{R_b - R_t}{H}\right) \cdot H_N. \tag{1}$$

Because $R_t = 0.000$ m in the 4-D maize model, in order to use the growth equations of the 4-D maize model, we assumed there were little variations of radius between the top stem and at the location of the stem to which the second leaf is attached (0.015 m from the top of the stem), namely, $R_t = 0.002$ m. Hereafter, in order to distinguish with the DLAmaize model, we named the model by coupling ELSYS, 4-D maize growth, and leaf break equations as the ELSYS coupling 4-D maize ($\text{ELSYS}_{4\text{Dmaize}}$) model. However, establishing a relationship between stem radius and H is still essential.

B. Dynamic L-System-Based Architectural Maize Model

We developed the DLAmaize model by integrating the modified growth equations of the 4-D maize model [50] with the ELSYS maize model to simulate 4-D maize structures. This modification concerned two parts: replacing the stem radius-leaf order function with a stem radius-height function [see (1)] and employing linear interpolation to transform the leaf width-length ratio from a constant value to a function that varies with N.

1) Relating the Stem Radius and Height: Considering that plants grew through cell division in an exponential manner owing to allometric scaling [61], [62], we used an exponential equation to link P_b and H, rather than a constant value of 0.008 in the 4-D maize model. The exponential equation was also used to describe P_t and H, where P_t was the ratio of stem top radius (R_t) and height (H)

$$\begin{cases} P_b = \dfrac{R_b}{H} = a_b \cdot e^{-\frac{H}{b}} = a_b \cdot e^{-\frac{H}{c \cdot H_{\max}}} \\ P_t = \dfrac{R_t}{H} = a_t \cdot e^{-\frac{H}{b}} = a_t \cdot e^{-\frac{H}{c \cdot H_{\max}}} \end{cases} \tag{2}$$

with a_b and a_t the maximum bottom and top stem radius to height, respectively. b was the maize height at which the stem radius changed from increasing to decreasing with H to simulate the decrease of stem radius with maize aging. It was the product of the stem maximum height $H_{\max}$ by the crop type-dependent parameter c. Equation (2) led to expressions of R_b (and R_t)

$$\begin{cases} R_b = H \cdot P_b = a_b \cdot H \cdot e^{-\frac{H}{b}} = a_b \cdot H \cdot e^{-\frac{H}{c \cdot H_{\max}}} \\ R_t = H \cdot P_t = a_t \cdot H \cdot e^{-\frac{H}{b}} = a_t \cdot H \cdot e^{-\frac{H}{c \cdot H_{\max}}}. \end{cases} \tag{3}$$

By taking the derivative of Equation (3), we can get the value of H when the radius started to decay

$$\begin{cases} R_b' = \dfrac{-a_b \cdot (H - c \cdot H_{\max}) \cdot e^{-\frac{H}{c \cdot H_{\max}}}}{c \cdot H_{\max}} \\ R_t' = \dfrac{-a_t \cdot (H - c \cdot H_{\max}) \cdot e^{-\frac{H}{c \cdot H_{\max}}}}{c \cdot H_{\max}}. \end{cases} \tag{4}$$

By setting R_b' and R_t' to 0, we got

$$\begin{cases} R_b = \max(R_b) & , \text{if } H = c \cdot H_{\max} \\ R_t = \max(R_t) & , \text{if } H = c \cdot H_{\max}. \end{cases} \tag{5}$$

When H was less than $c \cdot H_{\max}$, R_b (and R_t) increased with H, and when H was greater than $c \cdot H_{\max}$, R_b (and R_t) decreased with H. Typically, R_N was calculated using R_t and R_b with Equation (1).

We hypothesized that a linear relationship could approximate the relationship between a_b and a_t

$$a_t = d \cdot a_b \tag{6}$$

with d a ratio of a_t to a_b. Equation (6) could calculate a_t from a_b, and vice versa. The value range of a, c, and d should be larger than 0 and smaller than 1.

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2) Relating Leaf Width-Length Ratio and Leaf Order: The 4-D maize model [50] sets the leaf width-length ratio ($R_{W,L}$) at a constant value of 0.35 (A.10). Since maize has no secondary tissue growth, its leaf shape at any growth phase is fully developed and truncated horizontally in the cell elongation zone, and the relationship between leaf width and length is assumed to be linear [37]. As leaf length at different plant heights could vary significantly with water conditions [63], $R_{W,L}$ were allowed to vary vertically. We divided the maize plant into three layers (i.e., top, middle, and bottom), based on N. We selected two representative leaves per layer and measured their basal width and length to obtain $R_{W,L}$ for each layer. Based on the measured $R_{W,L}$ values in the three layers, we computed $R_{W,L}(N)$ of any order N using linear interpolation

$$\begin{cases} R_{W,L,N1} = \min(R_{W,L}) \\ R_{W,L,N_{\max}} = \max(R_{W,L}) \\ R_{W,L}(N) = R_{W,L,N1} + \dfrac{R_{W,L,N\max} - R_{W,L,N1}}{N_{\max} - 1} \cdot (N-1) \end{cases} \tag{7}$$

with $R_{W,L,N1}$ the ratio of the leaf with order N =1, and $R_{W,L,N\max}$ the ratio of the leaf with order $N_{\max}$.

C. Dynamic Heterogeneous Maize Canopy Modeling

A single maize plant was modeled using the DLAmaize model. Subsequently, the entire heterogeneous maize scene was simulated by replicating this plant at specific locations determined by plant density and distance between rows, with a random azimuth angle. As our focus was on a single-maize plant simulation, the DLAmaize model did not incorporate leaf orientation plasticity and the influence of each plant on others (i.e., competition for light). The parameters for the 4-D maize scene (Table II) were from the literature [64].

In addition, an equivalent homogeneous canopy was simulated to compare its reflectance with that of the 4-D maize architecture canopy. It was characterized by a spherical leaf angle distribution (LAD), commonly used to describe maize canopies [64], and the GAI from the heterogeneous scene (3-D maize scene) for each T_d. The values of GAI were inputted as leaf area index (LAI) into the homogeneous canopy as a replacement because of the missing corresponding parameter for 1-D models.

D. Radiative Transfer Modeling

Canopy reflectance was simulated with three RTMs.

1) *SAIL [54]:* It uses a discrete ordinate method with four fluxes (direct sunlight, flux to a sensor, and upward and downward hemispherical fluxes), and it treats vegetation as a 1-D vertical profile of the turbid medium.
2) *RGM [46]:* It uses a radiosity method (matrix of view factors between pairs of facets). It simulates 3-D vegetation as facets using the L-system. It is derived from thermal engineering and computer graphics.
3) *DART [13]:* It simulates the 3-D radiative budget and RS observations of Earth surfaces with a so-called flux tracking (FT) approach based on the "discrete ordinate" method with any number of discrete directions. It is among the most comprehensive RTMs. It is highly accurate and serves as a reference to validate the reflectances simulated by the RGM and SAIL models here.

Reflectance (R_{leaf}) and transmittance (T_{leaf}) of maize leaves at wavelengths 660 (red) and 880 nm [near-infrared (NIR)] under consideration were from the leaf optical properties experiment 93 (LOPEX93: lo93r/t0171) database. Soil reflectance (R_{soil}) was from the study [64].

Fig. 3 illustrates the flowchart for the maize canopy reflectance simulation. The DLAmaize model utilized input parameters ($N_{\max}$, T_{d}, $S_{\max}$, $\theta_{\max}$, and $H_{\max}$) to calculate the geometric parameters of an individual maize plant at a given T_d, including N, L_N, W_N, H_N, θ_N, φ_N, R_t, and R_b. The leaf breakpoint position (P_{break}) was derived from N and L_N. P_{break}, N, L_N, W_N, H_N, θ_N, φ_N, R_t, and R_b were input into the ELSYS model to simulate a single-maize structure. Finally, the RTMs employed this single-maize structure, scene parameters (D, d_{rows}), sun-view parameters (sun zenith angle SZA, sun azimuth angle SAA, view zenith angle VZA, and view azimuth angle VAA), and canopy optical properties (R_{soil}, R_{leaf}, and T_{leaf}) to simulate the maize scene and its BRFs. Since the model had random quantities, we conducted five replication experiments, and their mean values were presented.

IV. Four-Dimensional Maize Scene Modeling

A. Single 4-D Maize Modeling Using the DLAmaize Model

1) Calibration of the DLAmaize Model Coefficients Using the Field-Measured Data: In order to calibrate coefficients between stem radius and height, (2) was validated using field-measured maize geometry data. We divided all measured data from the normal growth and stress-impacted groups into exponential fitting (70%) and evaluation (30%) groups. We first examined the relationship between P_b (or P_t) and H. Figs. 4 and 5 display the fitting results from the normal growth and stress-impacted groups, respectively.

In the normal growth group's modeling, P_b exhibited a negative correlation with H, with high P_b values for low H. As H increased (plant gradually grew), P_b decreased and gradually approached a constant value. In the mature period, $P_b \approx 0.008$. The coefficient of determination (R^2) and root-mean-squared error (RMSE) were used to evaluate the fitting accuracy. We employed (2) to fit the modeling data, achieving good accuracy for P_b ($R^2 = 0.9692$ and RMSE = 0.0008) and P_t ($R^2 = 0.9108$ and RMSE = 0.0010)

$$\begin{aligned} P_b &= 0.023 \cdot e^{-\frac{H}{1.803}} = 0.023 \cdot e^{-\frac{H}{0.620 \cdot H_{\max}}} \\ P_t &= 0.019 \cdot e^{-\frac{H}{1.803}} = 0.019 \cdot e^{-\frac{H}{0.620 \cdot H_{\max}}}. \end{aligned} \tag{8}$$

Applying the derived fitting (8) to the validation data yielded good fitting accuracy for P_b ($R^2 = 0.8670$ and RMSE = 0.0025) and P_t ($R^2 = 0.8254$ and RMSE = 0.0023). In addition, the c value equaled 0.620, and the d value equaled 0.819 from the normal growth group. Using the above-mentioned coefficients and (3), we got a relationship

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TABLE II
INPUT PARAMETERS OF THE MAIZE SCENE REFLECTANCE MODELING. NIR: NEAR-INFRARED

Parameters		Values	
Single maize			
N_{max}	Maximum number of leaves produced by one plant	22	
S_{max}	Maximum total leaf area produced by one plant (m^2)	0.5	
θ_{max}	Inclination of the biggest leaf (°)	45	
T_d	Degree day (°D)	100-1950 (Step 50)	
H_{max}	Maximum height of stem (m)	2	
Maize scenery			
D	Plant density (plants / m^2)	8.8	
d_{rows}	Distance between rows (m)	0.75	
Sun-target-view geometry			
SZA	Sun zenith angle (°)	45	
SAA	Sun azimuth angle (°)	315	
VZA	View zenith angle (°)	0, 45, 45	
VAA	View azimuth angle (°)	0, 315, 135	
Optical property		Red	NIR
R_{soil}	Soil reflectance (%)	13.6	21.6
R_{leaf}	Leaf reflectance (%)	8.48	46.58
T_{leaf}	Leaf transmittance (%)	0.62	47.59

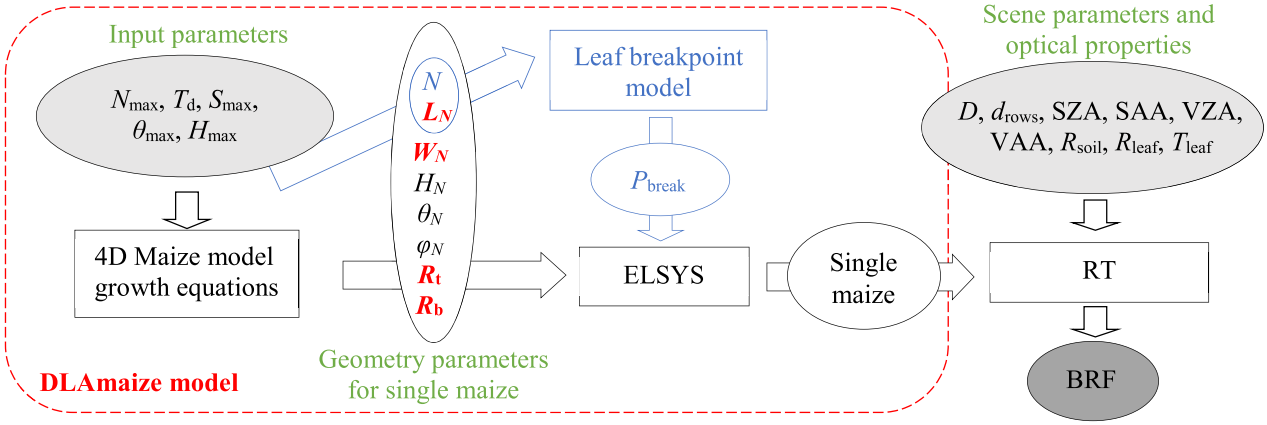


Fig. 3. Flowchart of the simulation of the BRF of the 4-D maize canopy using RT, with input parameters in lightly shaded ellipses, output parameters in darkly shaded ellipses, and intermediate parameters in white ellipses. Key parameters include N (leaf order), L_N (leaf length), W_N (leaf basal width), H_N (leaf vertical height), θ_N (leaf inclination angle), φ_N (leaf azimuth angle), R_t (maize top stem radius), and R_b (maize bottom stem radius). Parameters in red highlight differences between growth equations of the DLAmaize model and the 4-D maize model [50]. Parameters in blue are sourced from the maize leaf break model [27]. Descriptive texts are in green. Other parameters are in Table II.

between R_b (and R_t) and H in the normal growth group

$$R_b = H \cdot P_b = 0.023 \cdot H \cdot e^{-\frac{H}{0.620 \cdot H_{max}}}$$
$$R_t = H \cdot P_t = 0.019 \cdot H \cdot e^{-\frac{H}{0.620 \cdot H_{max}}}. \quad (9)$$

We validated (9) with field measurements [Fig. 6(a)]: $R^2 = 0.3459$ and RMSE $= 0.0017$ for R_b, and $R^2 = 0.4710$ and RMSE $= 0.0014$ for R_t.

In the stress-impacted group's modeling, P_b exhibited a negative correlation with H, with high P_b values for low H. As H increased (plant gradually grew), P_b decreased and gradually approached a constant value. In the mature period, $P_b \approx 0.008$. We employed (2) to fit the modeling data, obtaining good accuracy for P_b ($R^2 = 0.8428$ and RMSE $= 0.0057$) and P_t ($R^2 = 0.8762$ and RMSE $= 0.0052$)

$$P_b = 0.058 \cdot e^{-\frac{H}{0.695}} = 0.058 \cdot e^{-\frac{H}{0.267 \cdot H_{max}}}$$
$$P_t = 0.052 \cdot e^{-\frac{H}{0.695}} = 0.052 \cdot e^{-\frac{H}{0.267 \cdot H_{max}}}. \quad (10)$$

Applying the derived fitting (10) to the validation data yielded good fitting accuracy for P_b ($R^2 = 0.8139$ and RMSE $= 0.0063$) and P_t ($R^2 = 0.8166$ and RMSE $= 0.0067$). In addition, the c value equaled 0.267, and the d value equaled 0.896 from the stress-impacted group. We could also determine the relationship between R_b (and R_t) and H in the stress-impacted group [Fig. 6(b)]

$$R_b = H \cdot P_b = 0.058 \cdot H \cdot e^{-\frac{H}{0.267 \cdot H_{max}}}$$
$$R_t = H \cdot P_t = 0.052 \cdot H \cdot e^{-\frac{H}{0.267 \cdot H_{max}}} \quad (11)$$

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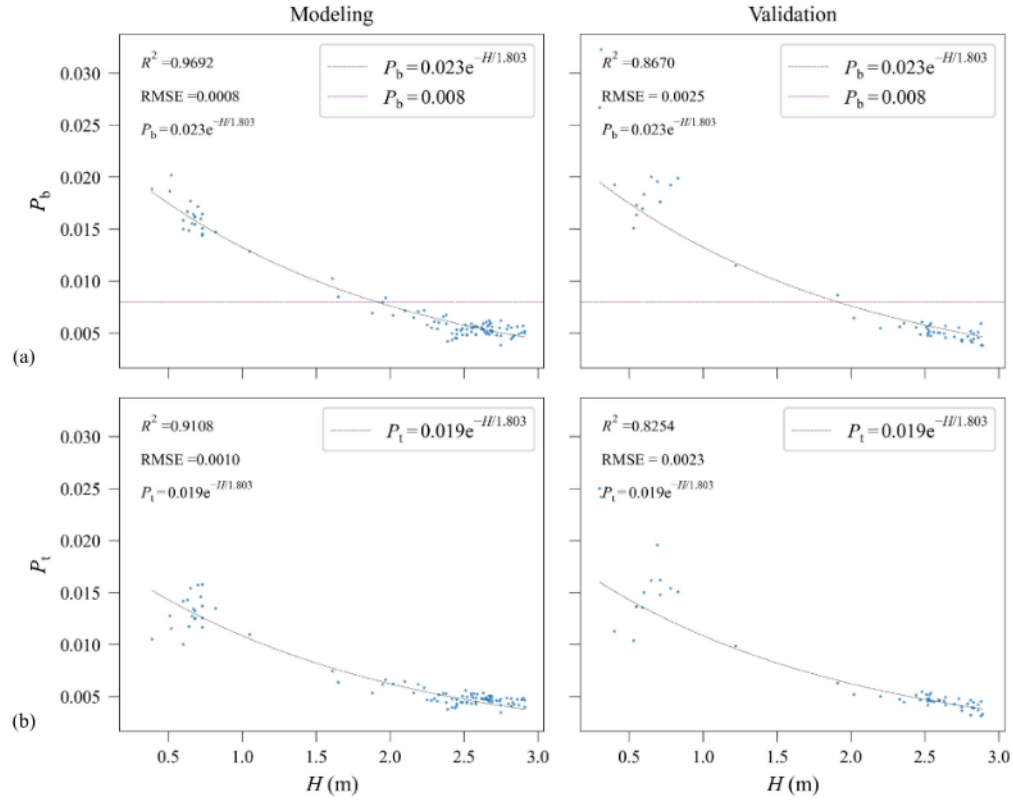


Fig. 4. Exponential relationship between (a) stem ratio (P_b) of bottom radius (R_b) and height (H) and between (b) stem ratio (P_t) of top radius (R_t) and H. Modeling (left) and validation (right) groups with data randomly assigned to these groups in a 7:3 ratio. Data are from the normal growth group.

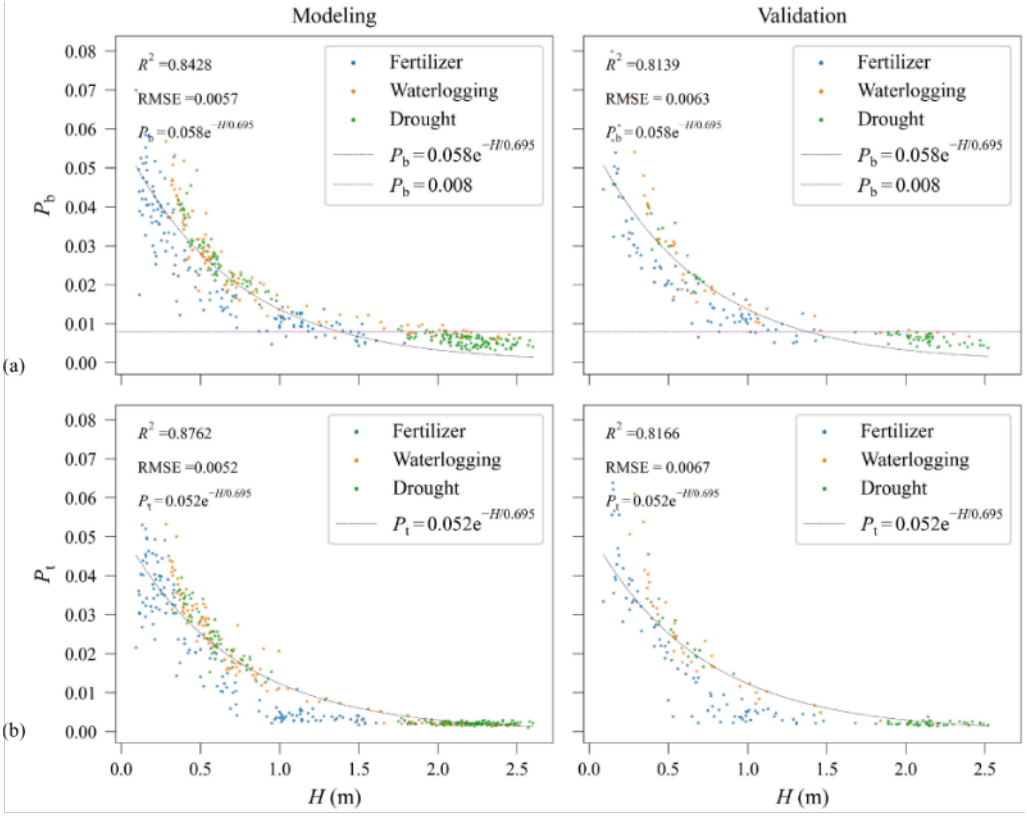


Fig. 5. Exponential relationship between (a) stem ratio (P_b) of bottom radius (R_b) and height (H) and between (b) stem ratio (P_t) of top radius (R_t) and H. Modeling (left) and validation (right) groups with data randomly assigned to these groups in a 7:3 ratio. Data are from the stress-impacted group. Colors distinguish the three experiments: fertilizer, waterlogging, and drought.

We validated (11) with field measurements [Fig. 6(b)]: $R^2 = -1.3980$ and RMSE $= 0.0051$ for R_b, and $R^2 = 0.3185$, and RMSE $= 0.0035$ for R_t. The separated fitting error metrics for each stress experiment are in Appendix B.

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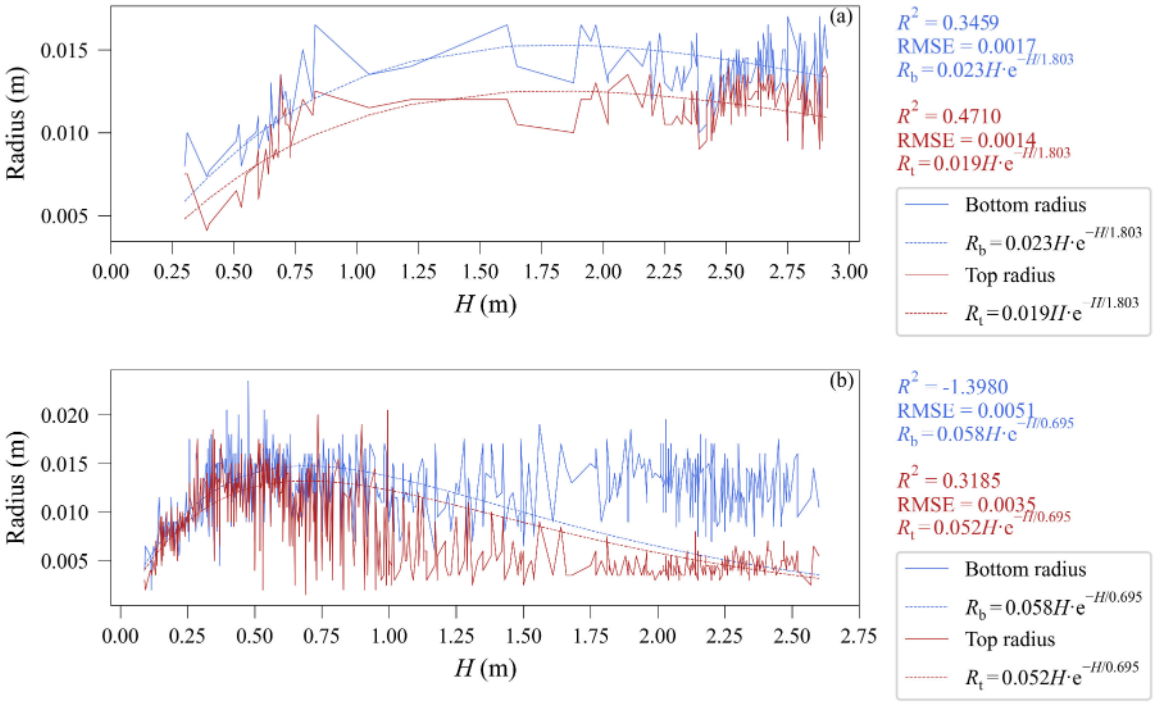


Fig. 6. Relationship between stem radius and height (H). (a) Normal growth group. (b) Stress-impacted group. Blue is for the bottom stem radius (R_b), and red is for the top stem radius (R_t). Solid lines represent the measured data, and dashed lines represent the trend line of the fitting equations.

Compared with the normal growth group ($c = 0.620$), the stress-impacted group exhibited a lower R^2 and a smaller c value ($c = 0.267$). The stress-impacted group appeared to reach the maximum stem radius earlier than the normal growth group. This phenomenon might occur because, under growth stress, maize nutrients were insufficient, causing the stem radius to enter attenuation early. Consequently, the relationship between H and R_b (or R_t) might better fit the normal growth maize in our case.

Only the stress-impacted group measured the leaf shape ratio $R_{W,L}$. The median value of $R_{W,L}$, derived from field measurements, exhibited a specific trend: upper canopy (0.36)> middle canopy (0.32)> lower canopy (0.29). In addition, no significant effects of waterlogging, drought, or fertilizer on $R_{W,L}$ were observed (Fig. 7).

2) Calibration of DLAmaize Model Coefficients With the 4-D Maize Model: We modified the growth equations of the 4-D maize model by establishing a relationship between $R_b(R_t)$ and H and replacing $R_{W,L}$ from a constant value (0.35) to a function with N, and proposed the DLAmaize model.

The first step of building the DLAmaize model was to fit the relationship between R_b and H. Based on (8), the c value (0.620) and d value (0.819) from the normal growth group and two constant values at the top ($H = 2.000$ m and $R_b = 0.016$ m) and bottom ($H = 0.015$ m and $R_b = 0.002$ m) from the 4-D maize model were used to fit a_b (0.040) and a_t (0.033) and the value boundary during fitting was limited within the normal growth group and stress-impacted group. We then obtained a new relationship between $R_b(R_t)$ and H, which could replace the relationship (A.9) in the 4-D maize model

$$R_b = H \cdot P_b = 0.040 \cdot H \cdot e^{-\frac{H}{0.620 \cdot H_{\max}}}$$
$$R_t = H \cdot P_t = 0.033 \cdot H \cdot e^{-\frac{H}{0.620 \cdot H_{\max}}}. \quad (12)$$

Fig. 8 compares the relationship between H and $R_b(R_t)$ in the 4-D maize model and DLAmaize model [see (12)], with other input parameters identical to those in the study [50]. It shows that, for the 4-D maize model, the relationship between R_b and H was nearly positively linear; however, for the DLAmaize model [see (12)], the maize stem's radius increased with height. As the maize stem grew, the maize radius growth rate slowed down and even decreased. Therefore, the pattern of the DLAmaize model aligned more closely with plant anisotropic growth.

The second step of building the DLAmaize model was to replace the constant $R_{W,L}$ value (0.35) in (A.3) with a function of N. The linear interpolation [see (7)] provided $R_{W,L}(N)$ with $R_{W,L,N1} = 0.29$ and $R_{W,L,N\max} = 0.36$, based on the field measurements

$$\begin{cases} R_{W,L,N1} = 0.29 \\ R_{W,L,N_{\max}} = 0.36 \\ R_{W,L}(N) = R_{W,L,N1} + \dfrac{R_{W,L,N_{\max}} - R_{W,L,N1}}{N_{\max} - 1} \cdot (N - 1). \end{cases} \quad (13)$$

B. Maize Growth Scene Modeling

Current plant modeling endeavors often run the risk of oversimplifying the depiction of maize stem and a simplification that may significantly impact the simulation of RT

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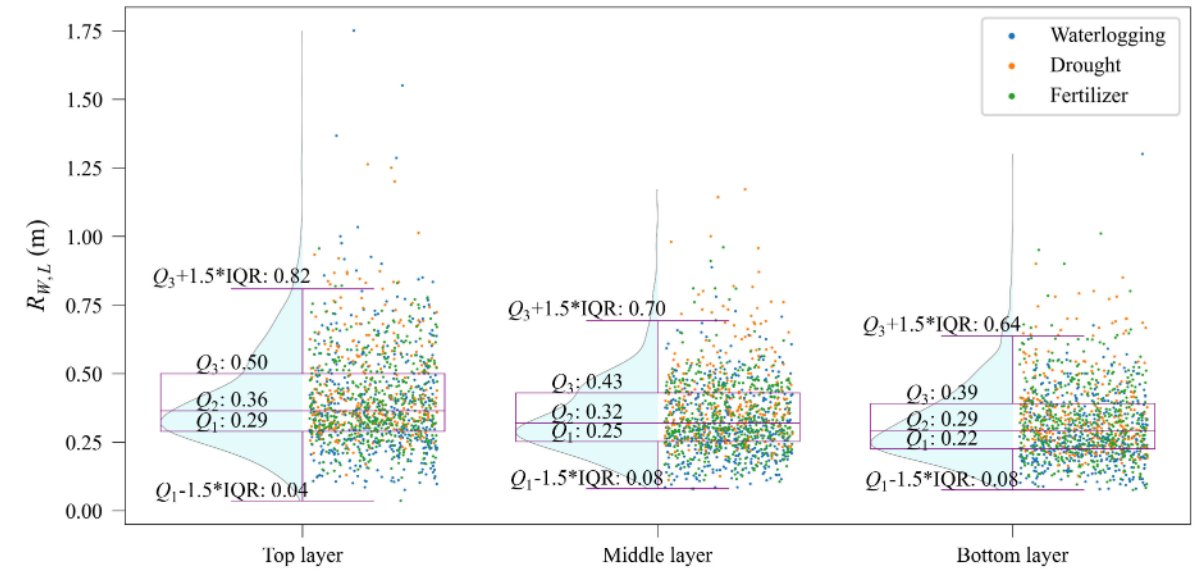


Fig. 7. Distribution of leaf width-length ratio ($R_{W,L}$) at the three layers (top, middle, and bottom). The figure is plotted using the box (left), strip (right), and violin (middle) plots, and quartiles Q_1, Q_2, and Q_3 per the box plot. The colors of the scatters distinguish the three stress-impacted experiments.

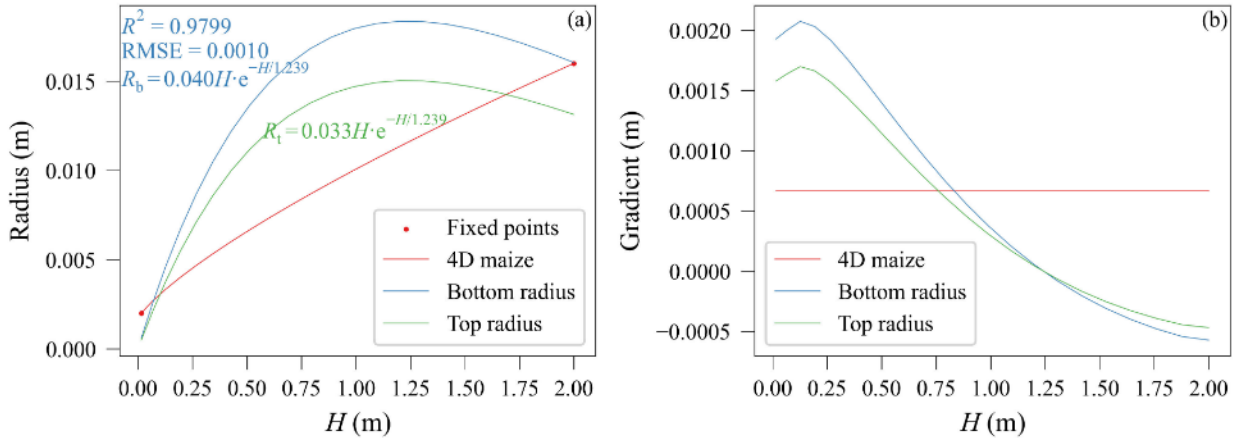


Fig. 8. (a) Relationship between stem radius (R_t: top radius and R_b: bottom radius) and height (H) and (b) corresponding gradient. The two red points (H =0.015 m and R_b =0.002 m. H =2.000 m and R_b =0.016 m) indicate the fixed values used for fitting. The red line indicates the near-linear fitting equation from the 4-D maize model. The blue curve indicates the bottom radius of the stem for the DLAmaize model. The green line indicates the top radius of the stem for the DLAmaize model. The coefficient of determination (R^2) and RMSE are calculated using the two red points (H =0.015 m and R_b =0.002 m. H =2.000 m and R_b =0.016 m) with R_b from 4-D maize model considered as true values and R_b from DLAmaize model considered as estimated values.

within the canopy [51]. Stem plays a significant role in the processes of canopy photosynthesis, respiration, and transpiration [42]. Canopy reflectance is intricately linked to the spatial arrangement of both stems and leaves. Moreover, the spatial distribution of stems indirectly influences the distribution of leaves within the canopy. Furthermore, plant stems can cast shadows upon the underlying surfaces, a phenomenon of considerable significance. Precisely modeling these shadows becomes imperative since they notably influence the amount of light that reaches the lower canopy layers [65]. An accurate representation of stem characteristics in models allows for capturing this inherent heterogeneity, leading to more faithful simulations of light distribution within the canopy and its interactions with various plant components.

Fig. 9 displays the maize 4-D scene by $\text{ELSYS}_{\text{4Dmaize}}$ [Fig. 9(a)] or DLAmaize [Fig. 9(b)] at six growth stages (T_d =300°D, 600°D, 900°D, 1200°D, 1500 °D, 1800 °D). In the DLAmaize model, we used the relationship (12) between R_t and H (and R_b and H), along with the linear interpolation (13) that provided $R_{W,L}(N)$. The primary differences between the two models were stem radius and leaf width and length. These differences directly led to variations in canopy structure parameters, such as the foliage cover (FC) and GAI.

Fig. 10 provides an insight into the variation of FC and GAI across a range of T_d, spanning from 100 °D to 1950 °D, with increments of 50 °D. The maize scene exhibited specific GAI and FC values at each developmental stage. In the context of a heterogeneous canopy, the DART model calculated the FC of the 3-D scene as the vertical projection of maize plants. Conversely, FC was computed as exp(−GAI) for a homogeneous canopy. The observed growth pattern revealed that maize underwent significant growth from 100 °D to

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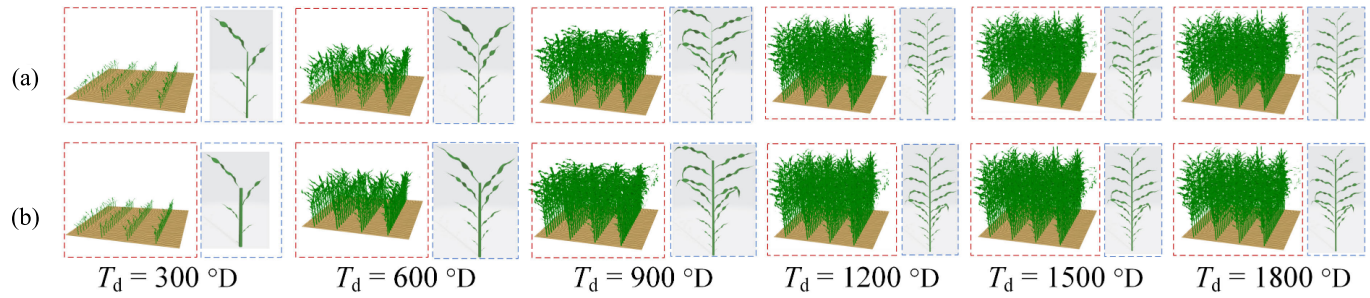


Fig. 9. Four-dimensional maize scene at six-degree days (T_d) from emergence to male anthesis with withering simulated by the gradual loss of bottom leaves. (a) $ELSYS_{4Dmaize}$ model. (b) DLAmaize model. The main differences between them are stem radius, leaf width, and length. The scenes are displayed using the DART model. The red dashed box denotes a 3-D maize scene with identical scaling for growth scene comparison, whereas the blue dashed box highlights a 3-D single-plant maize figure with different scaling to accentuate finer details.

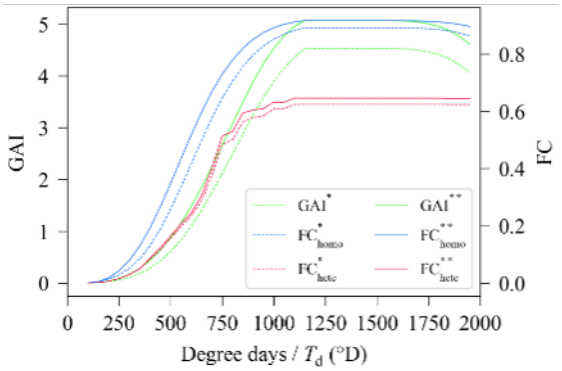


Fig. 10. Variation of GAI (left y-axis) and FC (right y-axis) with degree days (T_d) plots. The FC of the homogeneous (FC_{homo}) and heterogeneous (FC_{hete}) canopies significantly differ, particularly for large GAI. Dash lines (*): $ELSYS_{4Dmaize}$. Solid lines (**): DLAmaize model.

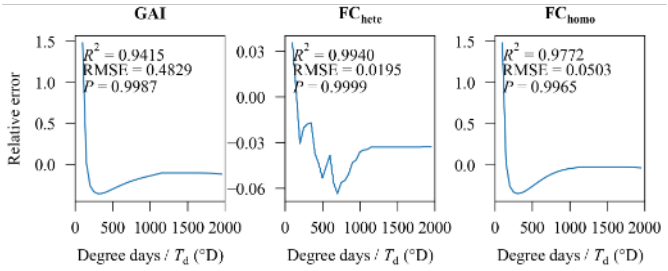


Fig. 11. Relative error in GAI, heterogeneous FC (FC_{hete}), and homogeneous FC (FC_{homo}) of the simulated maize scene between $ELSYS_{4Dmaize}$ and DLAmaize. The values from DLAmaize are considered true values and those from $ELSYS_{4Dmaize}$ are considered estimated values. The coefficient of determination (R^2), RMSE, and the standard correlation coefficient (P) are calculated using GAI or FC, rather than relative error during the full growth stage. P is calculated using the Pearson method.

1150 °D, resulting in substantial increases in GAI and FC. Subsequently, a stable stage was attained, followed by a decline in FC and GAI for T_d values exceeding 1550 °D. The absolute value of maximum differences between the DLAmaize and $ELSYS_{4Dmaize}$ were observed to be 0.6461 (for GAI) and 0.0302 (for FC). Even though the homogeneous and heterogeneous canopies exhibited equivalent GAI at each growth stage, their FC values could exhibit significant disparities. The absolute value of maximum FC difference between homogeneous and heterogeneous canopies when employing the DLAmaize model was recorded as 0.2915.

We calculated the relative error in GAI [Fig. 11(a)], FC heterogeneous [Fig. 11(b)], and FC homogeneous [Fig. 11(c)] of the simulated maize scene between $ELSYS_{4Dmaize}$ and DLAmaize. The values from DLAmaize were considered true values, and those from $ELSYS_{4Dmaize}$ were considered estimated values. $ELSYS_{4Dmaize}$ overestimated GAI and FC in the early stage of maize growth, and this overestimation gradually decreased and even became an underestimation. In the later growth stage, $ELSYS_{4Dmaize}$ and DLAmaize gradually converged and stabilized. The reason for the overestimation of GAI by $ELSYS_{4Dmaize}$ in the early stage was that a fixed leaf width-length ratio was used, whereas the ratio of DLAmaize was slightly smaller than that of $ELSYS_{4Dmaize}$ at this time, resulting in a slightly smaller GAI than $ELSYS_{4Dmaize}$. As T_d progressed, the stem radius and leaf width-length ratio of DLAmaize gradually increased, so GAI gradually exceeded $ELSYS_{4Dmaize}$. In the later growth stage, GAI values were large and stable, so the relative error also tended to be small and stable. FC homogeneous was calculated based on GAI, so it showed a similar trend with GAI. Owing to the clumping effect [66], FC heterogeneous curves were not as smooth as GAI, but the trend was consistent.

We further calculated the relative error in FC between homogeneous scene and heterogeneous scene with simulated maize using $ELSYS_{4Dmaize}$ [Fig. 12(a)] and DLAmaize [Fig. 12(b)]. Owing to the clumping effect, homogeneous FC was always larger than heterogeneous FC. However, the magnitude varied, particularly in the early growth stage when FC was small.

V. Reflectance Simulation of the Maize Scene

A. Time Series Maize Canopy BRF Simulation

The time series scene BRFs were simulated with the 4-D canopy created by $ELSYS_{4Dmaize}$ [Fig. 9(a)] and DLAmaize [Fig. 9(b)] models. The T_d evolution of the BRFs of the homogeneous scenes (SAIL and DART homogeneous) and 3-D scenes (RGM and DART heterogeneous) are displayed in red [Figs. 13(a) and 14(a)] and NIR [Figs. 13(b) and 14(b)] bands for the nadir, hotspot (HS), and darkspot (DS) viewing directions. These two bands and three directions were considered to emphasize the influence of the canopy structure.

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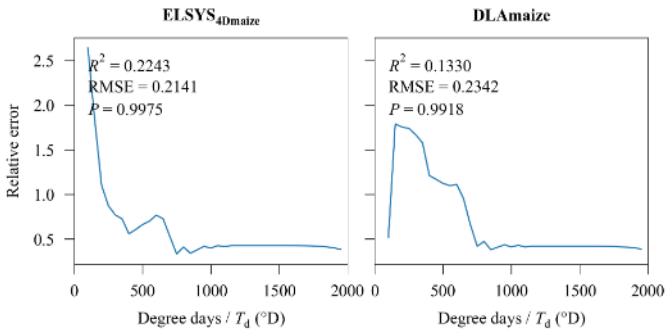


Fig. 12. Relative error in FC between homogeneous scene and heterogeneous scene with simulated maize by $ELSYS_{4Dmaize}$ and DLAmaize. The values from the heterogeneous scene are considered true values, and those from the homogeneous scene are considered estimated values. The coefficient of determination (R^2), RMSE, and the standard correlation coefficient (P) are calculated using FC, rather than relative error during the full growth stage. P is calculated using the Pearson method.

The BRFs of the heterogeneous scenes fluctuated during the stable period owing to the fluctuating FC of these scenes. BRFs of all scenes were the highest in the HS direction in both red and NIR bands. They were the lowest in the DS direction for the red band and the nadir direction for the NIR band. The DS reflectance was higher than the nadir reflectance in the NIR band, contrary to the red band, which confirmed the statement [10], [67], [68] that reflectance was not always lowest in the DS direction, particularly in the NIR band with strong multiple scattering. It highlighted the significant impact of canopy structure on the BRF anisotropy of maize fields.

B. Cross-Model Validation for Reflectance Simulation

Fig. 15 shows a comparative analysis of the BRFs simulated by the SAIL model and RGM in relation to those simulated by the DART model through the integration of the $ELSYS_{4Dmaize}$ and DLAmaize models. The overall findings exhibited a remarkable level of consistency. In the case of homogeneous scenarios, denoted as "SAIL-DART," and heterogeneous scenarios, denoted as "RGM-DART," the R^2 consistently hovered approximately 0.9 across most instances, with negligible deviations. The higher degree of BRF consistency observed in the red band, as opposed to the NIR band, can be attributed to the robust multiple scattering effects inherent in the NIR band, along with variations inherent to the RT algorithms of models. Furthermore, the agreement in BRFs appeared to be more pronounced for homogeneous scenes ("SAIL-DART") than for heterogeneous scenes ("RGM-DART").

The significant disparity in high reflectances, evident in the comparative analysis of DART to SAIL models in all cases in the NIR band, could be attributed to the different calculations of multiple scattering. DART and SAIL used the same equations for computing the first-order reflectance of homogeneous Lambert leaf elements, resulting in a good constituency in the red band. However, the NIR band had much stronger multiple scattering than the red band due to high reflectance. The differences between the DART and SAIL models for multiple-scattering calculations resulted in this significant disparity.

The notable distinctions observed in the NIR band in the comparative analysis of the DART model to RGM in the nadir direction, in contrast to the other viewing geometries. In addition, Figs. 13 and 14 show that the nadir direction in the NIR band exhibited fewer smooth transitions than the curves in other viewing directions. In the nadir observation direction, there was an interplay of both illuminated and shadowed regions, distinct from the HS (scenes predominated by illuminated region) and DS (scenes predominated by shadowed region). This might be because an increase in GAI led to an increasing NIR reflectance, particularly owing to the pronounced multiple scattering component of the NIR band. However, it also led to decreased NIR canopy reflectance due to the increased presence of shadows in the field of view. Therefore, the reflectance variations in the nadir direction were more intricate and complex than other observation directions. This may also explain why compared with DS and HS, the nadir direction witnessed notable simulated reflectance distinctions between the DART model and RGM in the red band. RGM and DART model used different calculations for multiple scattering and multiple scattering in the red band may not be so strong compared with NIR. Therefore, the simulated reflectances between the DART model and the RGM were closer in the red band than in the NIR.

To compare the simulated canopy reflectance between DLAmaize and $ELSYS_{4Dmaize}$, the reflectances generated by the DLAmaize model were considered the true values, whereas those produced by the $ELSYS_{4Dmaize}$ model were considered the measured values for calculating the relative error. As depicted in Fig. 16, the most significant differences were observed in the HS direction (the absolute value of the relative difference can be reached at most 67.4% in the red band), likely attributable to the leaf size variations owing to the differences in $R_{W,L}$ between the two models. It is well-documented that leaf size significantly influences canopy reflectance in the HS direction because it affects the proportion of shadows in the scene [69]. This shadow effect could also explain why the DS direction exhibited the second-largest differences. The nadir observation direction was the least influenced by these differences. This was largely owing to the relatively insignificant variations in GAI and FC between the two models.

C. Comparison of SAIL Model and RGM Simulated Reflectance

Fig. 17 comprehensively compares the BRFs simulated by the RGM for heterogeneous scenes and the SAIL model for homogeneous scenes. This comparison was carried out within the context of the $ELSYS_{4Dmaize}$ and DLAmaize models. In most cases, a high degree of correlation was observed, with R^2 values approximately equal to 0.9 in the majority of cases in the red band. The differences in BRFs were found to be most minimal in the HS direction, followed by the DS, whereas the nadir direction exhibited the largest discrepancies. Notably, in the red band, BRFs from homogeneous scenes consistently underestimated those from heterogeneous scenes across various viewing geometries in most instances. Conversely, in the NIR spectral domain, BRFs from homogeneous

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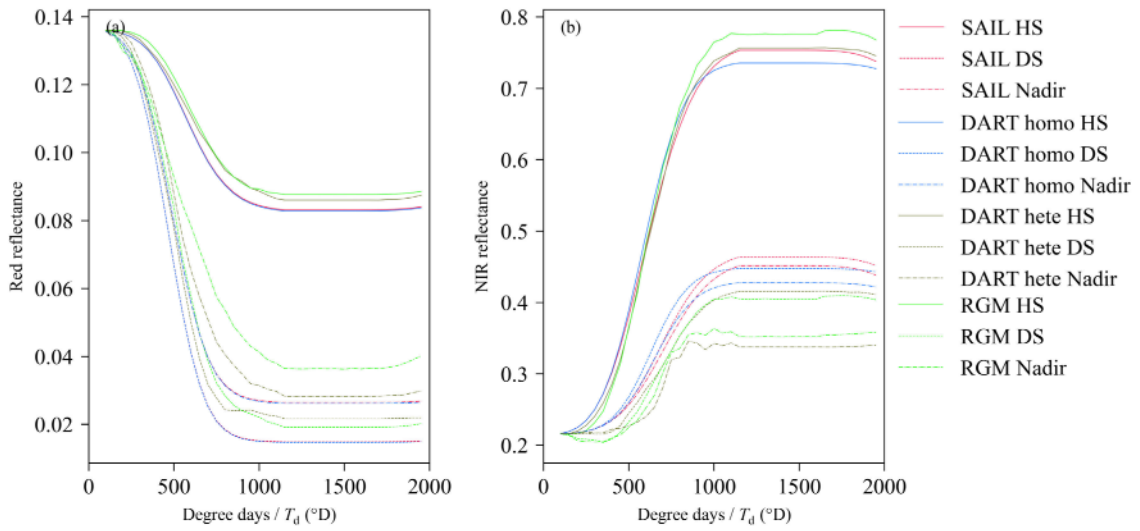


Fig. 13. ELSYS$_{4Dmaize}$ model: BRFs in (a) red and (b) NIR bands in the HS, DS, and nadir directions for the homogeneous and heterogeneous scenes as a function of degree days (T_d). BRF is simulated for homogeneous scenes by the SAIL model, for heterogeneous scenes by RGM, and for homogeneous and heterogeneous scenes by the DART model.

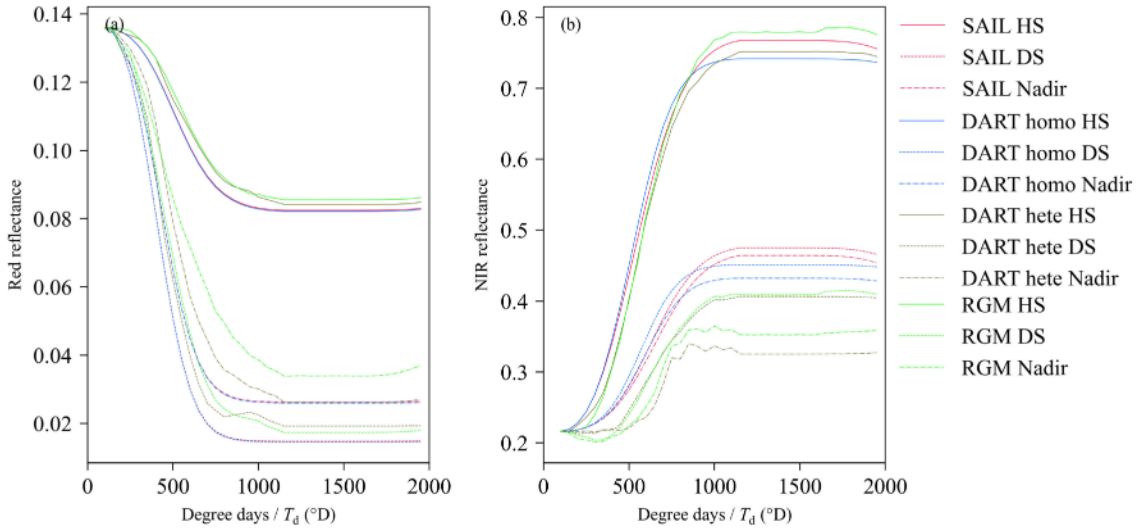


Fig. 14. DLAmaize model: BRFs in (a) red and (b) NIR bands in the HS, DS, and nadir directions for the homogeneous and heterogeneous scenes as a function of degree days (T_d). BRF is simulated for homogeneous scenes by the SAIL model, for heterogeneous scenes by RGM, and for homogeneous and heterogeneous scenes by the DART model.

scenes tended to overestimate those from heterogeneous scenes in most cases. The most pronounced differences in BRFs were observed along the nadir direction. These trends aligned with findings reported by Espana et al. [27]. Furthermore, the differences in BRFs simulated from homogeneous and heterogeneous canopies appeared to be more pronounced when red band reflectance was low and NIR band reflectance was high. This pattern was associated with high values of GAI and FC and was likely influenced by clumping effects. The leaves of maize tended to cluster together, resulting in a lower FC of heterogeneous canopies than that of homogeneous canopies.

The relative error of canopy reflectance simulated between homogeneous and heterogeneous scenes was used to validate the homogeneous assumption (Fig. 18). This assessment was based on a comparison between the reflectances generated by the SAIL model, considered as estimated values and the reflectances produced by the RGM, considered as true values. Notably, the nadir observation direction exhibited the most significant relative error between homogeneous and heterogeneous scenes, particularly the absolute value of the relative error can be up to 41.16% in the red band, and the HS direction demonstrated the lowest relative error within the red and NIR spectral bands. This suggests that the homogeneity assumption exerted a comparatively minor influence on observations in the HS direction but exerted a more substantial impact on those in the nadir direction. This effect could likely be attributed to clumping effects, resulting in a smaller proportion of soil being observed from the nadir direction for the homogeneous canopy than that of the heterogeneous canopy.

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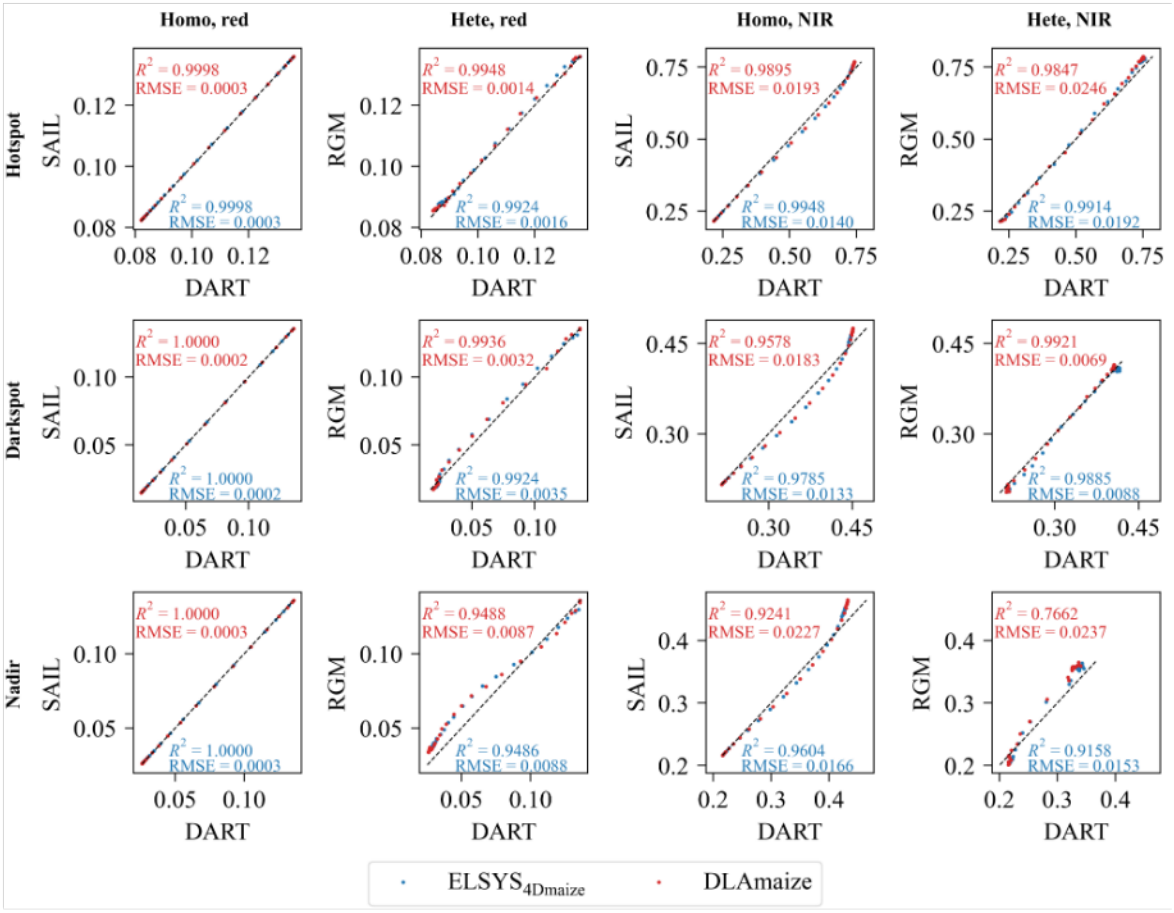


Fig. 15. SAIL BRFs of the homogeneous scenes (Homo) and RGM BRFs of the heterogeneous scenes (Hete) compared with DART model BRFs for the red and NIR bands and three directions (HS, DS, and nadir).

In terms of optical property, the overall trend indicated an underestimation of canopy reflectance in the red band and an overestimation in the NIR band for homogeneous assumption. This phenomenon might be linked to the overestimation of FC in the homogeneous scene. Furthermore, the relative error displayed variation with the growth stages of maize, following an overall pattern of increase, followed by a decrease, and subsequently, another increase and slight decrease. There were minimal differences in comparing the relative error of the homogeneous and heterogeneous scenes between the $ELSYS_{4Dmaize}$ and DLAmaize models.

VI. Discussion

A. Stem Radius-Height Ratio and Leaf Width-Length Ratio

The allometric growth rate of maize stems is characterized by a near-linear increase in height and a gradual slowing of the increase in radius with T_d. Specifically, in the early growth stages, the stem expands rapidly owing to the surface location of the growing point, swift root development and spread, and rapid stem elongation, followed by the development of the stem radius. As the plant matures, the increase in stem radius slows down. At this stage, the plant requires large amounts of nutrients for the distranslocation to the prospective male flower and the initiation of cob development [70]. Consistent with the literature [70], we observe that both P_b and P_t exhibit a negative relationship with maize stem height, eventually converging to a fixed value (Figs. 4 and 5). Furthermore, at the mature stage of maize (the identical stage when geometry parameters were measured in the study [50]), the values derived from our study closely align with the constant value (0.008) reported in the study [50].

Although the relationship between stem height and radius is not necessarily positive [71], this factor is seldom considered in current 4-D maize structure simulations. Presently, calculations of maize stem radius primarily rely on leaf order, typically employing a linear decrease of diameter with order [37], [50]. However, the DLAmaize model incorporates a nonlinear relationship. In addition, the Pearl growth equation is widely used to simulate the stem height and diameter [72], and this also indirectly demonstrates the feasibility of using an exponential relationship to fit the stem radius-height ratio with height.

We assume a constant leaf width-length ratio for each leaf, as maize lacks secondary tissue growth, and the leaf shape at any growth phase is fully developed and truncated horizontally in the cell elongation zone [37]. An earlier study [63] reveals that the sensitivity of leaf length to water varies at different vertical heights within the canopy. Consequently, we measured leaf ratios at various heights and determined

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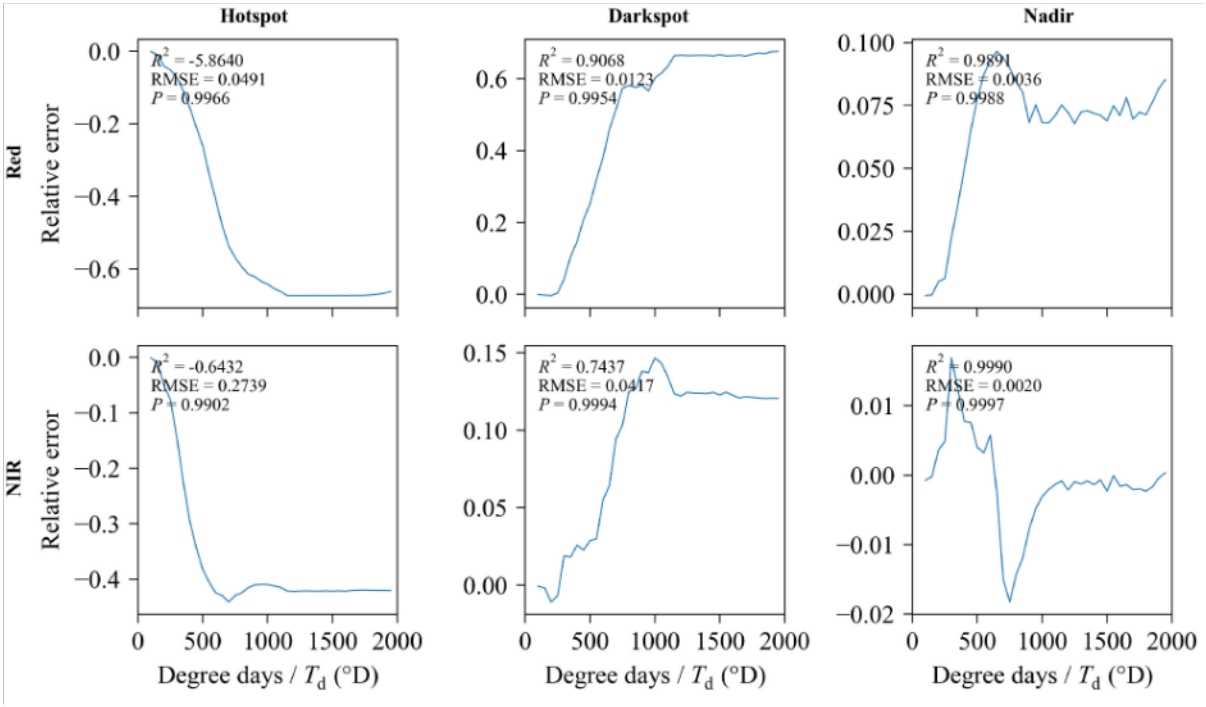


Fig. 16. The relative error in simulated maize canopy BRFs is determined by considering the reflectances generated by the DLAmaize model as the truth values, whereas those produced by the $ELSYS_{4Dmaize}$ are considered as the estimated values. The coefficient of determination (R^2), RMSE, and the standard correlation coefficient (P) are calculated using canopy reflectance, rather than relative error during the full growth stage. P is calculated using the Pearson method.

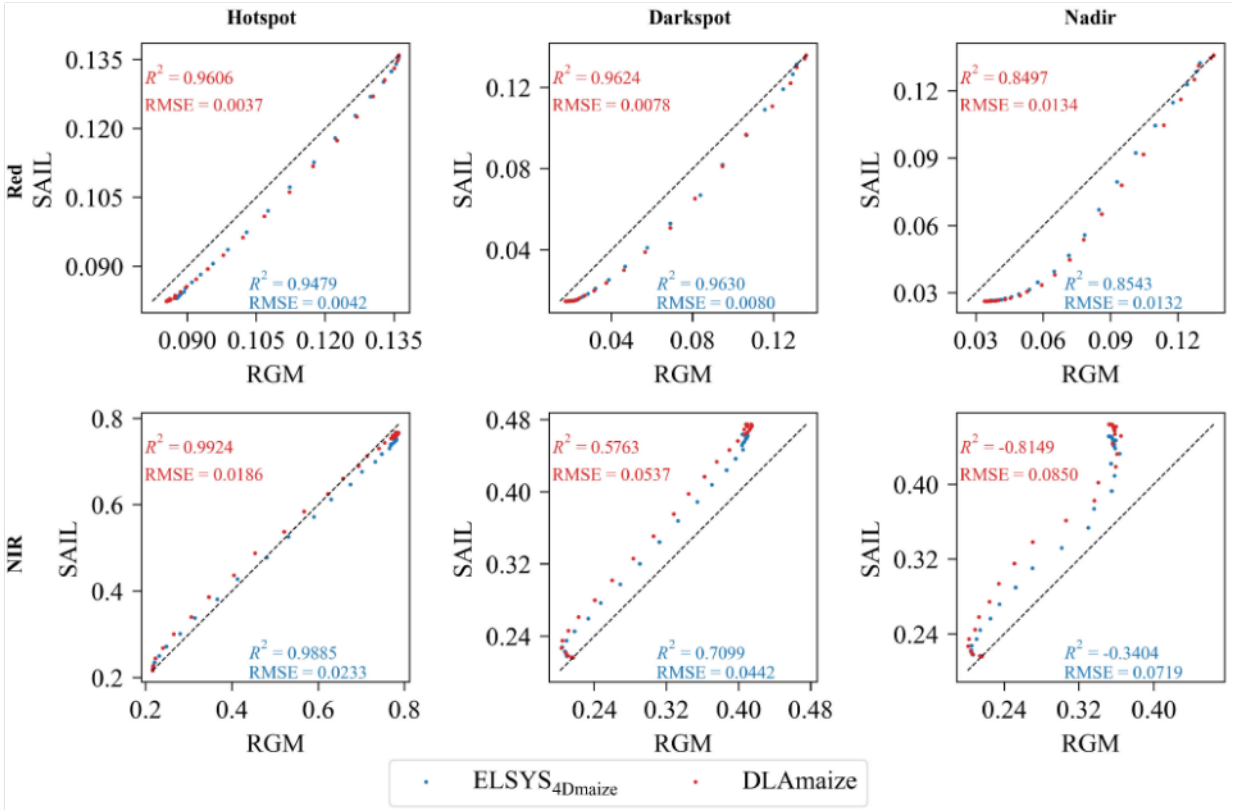


Fig. 17. Scattering by SAIL BRFs of homogeneous scenes versus RGM BRFs of heterogeneous scenes, for the red and NIR bands, in the HS, DS, and nadir viewing directions.

the value for each leaf order through linear interpolation (Fig. 7). Prior research has demonstrated that leaf width and stem diameter are more sensitive to nutritional status than leaf length [73], [74]. Despite this, the equation we derived

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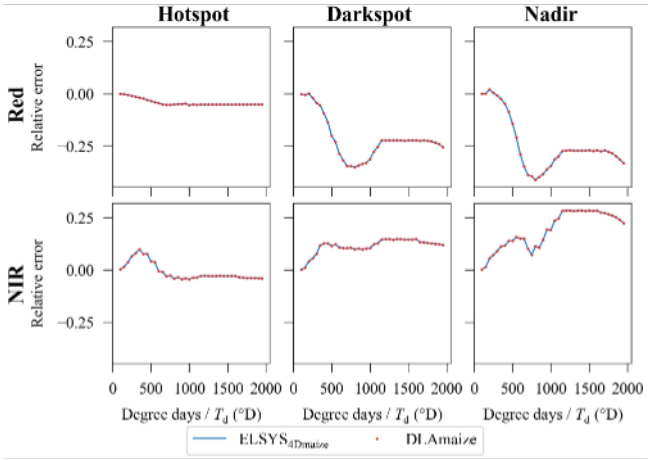


Fig. 18. Relative error of the simulated maize canopy BRFs between homogeneous and heterogeneous scenes by considering the reflectances of the SAIL model as estimated values and reflectances of RGM as true values.

was a fitting result under numerous fertilizer conditions, indicating that our conclusion is applicable to a wide range of situations.

By using RT simulations, main differences in canopy reflectance are observed between DLAmaize and $\text{ELSYS}_{4\text{Dmaize}}$ simulations, particularly in the HS direction. The main reason for the difference may be that we modify the $R_{W,L}$, which, in turn, changes the leaf size and the canopy reflectance in the HS direction is strongly affected by the leaf size [69]. Therefore, our modifications to the geometric structure of maize are of interest to canopy BRF simulations.

B. Four-Dimensional Maize Canopy Structure and Reflectance Simulation

Plant cover architecture is greatly oversimplified in most RTMs for practical applications in terms of computation time and computer memory. It is an essential source of inaccuracy in BRF simulation [27], [51]. For instance, an SAIL-based RTM struggles to handle severe leaf clumping in row crops like maize when simulating satellite Sentinel-2 data [75]. We observe a significant difference in FC between homogeneous and heterogeneous maize canopies with the same GAI, particularly for fully developed plants (high GAI) (Fig. 10). Typically, assuming homogeneity results in an overestimation of FC, particularly when GAI is large. Consequently, our study emphasizes the importance of employing accurate 3-D canopy modeling to simulate canopy reflectance.

Simplifying the 3-D canopy architecture can also impact the simulation of RT at the canopy's upper boundary. Therefore, using the exact 3-D structure of the canopy can be important for maize canopy reflectance simulations. There are significant differences between homogeneous and heterogeneous canopies in red and NIR bands [Figs. 13, 14, and 18]. For example, the nadir reflectance of the homogeneous canopy is considerably higher than that of the heterogeneous maize canopy in the NIR band and lower than that in the red band. Espana et al. [27] acknowledge this issue and attribute it to the row effect. Our findings indicate that simulating a vegetation canopy as a homogeneous turbid volume to estimate its BRF introduces significant inaccuracies. This underscores the necessity of considering the 3-D structure of vegetation canopies and the location of plants for BRF simulation.

Nadir direction tends to produce significant differences among RTMs. In the nadir observation direction, there is an interplay of illuminated and shadowed regions, distinct from the HS (scenes predominated by illuminated region) and DS (scenes predominated by shadowed region). Moreover, different models typically have widely differing mechanisms of illuminated and shadowed parts (particularly, between GO and RTMs), leading to significant differences in simulated canopy reflectances in the nadir observations.

VII. Conclusion

We aim to fulfill the time series simulation ability of the maize model in ELSYS [45] from emergence to male anthesis by considering architectural characteristics throughout the entire vegetative growing season using two methods. The first method directly couples the maize growth [50] and leaf break [27] equations with ELSYS, and is named $\text{ELSYS}_{4\text{Dmaize}}$. The second method modifies the growth equations in the 4-D maize model [50], including replacing the stem radius-leaf order function with a stem radius-height function and employing linear interpolation to transform the $R_{W,L}$ from a constant value to a function that varies with leaf order, thereby simulating the 4-D maize structure, and is named as DLAmaize model. Based on the field-measured maize geometry data, we show that the modified growth equations are able to describe the structural parameters of maize more accurately.

We import the 4-D maize scene into the RGM and DART models separately to simulate the canopy BRF, achieving good consistency. The leaf size and stem radius significantly affect the simulated canopy BRFs, particularly along the HS direction. In addition, we conduct a comparison between 3-D and 1-D models for canopy BRF simulation. Some general conclusions are as follows.

1) An exponential relationship can describe the stem radius-height ratio with height, which decreases as stem height increases. The leaf width-length ratio increases with leaf order.
2) Reflectance in the DS direction is not necessarily the lowest among all observation directions. In our case, for a maize scene (i.e., row-planted vegetation), reflectance is minimal in the DS direction in the red band and is minimal in the nadir direction in the NIR band.
3) Assuming a homogeneous canopy can result in significant errors, especially along the nadir observation direction. This emphasizes the importance of accurately simulating 3-D plant architecture and location.

Our study contributes to the use of maize growth models for simulating RS observations. The code of the DLAmaize model is distributed together with the DART model (https://dart.omp.eu/#/getDart, accessed on June 19, 2023). However, the DLAmaize model may be suitable for normal growth maize, and adjustments to the growth equations should be contemplated to enable the simulation of stress-impacted growth in maize. Since maize is characterized by allometric growth, encrypted measurements at the jointing stage can

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result in more evenly distributed morphological data. In addition, geometric parameters of plant height measured by LiDAR could serve as input parameters, potentially replacing the T_d to achieve a more accurate reconstruction of 3-D maize plant structure. It may present an avenue for future research.

APPENDIX A

For a leaf with order N, the leaf area (S_N) is given by [50]

$$S_N = \frac{1}{2}\left(1 + \sin\left(\frac{\pi}{0.59526} \cdot \left(\frac{N}{N_{\max}}\right)^{1.4158}\right) - \frac{\pi}{2}\right) \quad \text{(A.1)}$$

with $N_{\max}$, the maximum number of a single-maize leaf, specified by the user as a model input parameter. Since S_N of different varieties of maize varies considerably, it needs to be normalized based on the user input parameters as S'_N

$$S'_N = \frac{S_N \cdot S_{\max}}{\text{sum}(S_N)} \quad \text{(A.2)}$$

with $S_{\max}$ the maximum total leaf area produced by one plant, which the user specifies as a model input parameter. The basal width of the leaf (W_N) is

$$W_N = \sqrt{S'_N} \cdot R_{\text{W,L}} \quad \text{(A.3)}$$

with $R_{W,L}$ the leaf width-length ratio, a user-defined input (set as 0.35 in this case). Consequently, the leaf length is

$$L_N = \frac{\sqrt{S'_N}}{R_{W,L}}. \quad \text{(A.4)}$$

Depending on N, some maize leaves may have breakpoints. The leaf has a 15% probability of breaking if the leaf number is less than 12 and a 25% probability if the leaf number is greater than or equal to 12. If the leaf breaks, the break is randomly located between 0.25 and 0.95 times L_N, and the probability distribution follows a uniform distribution [27].

Leaf inclination angles (θ_N) are

$$\theta_N = S''_N \cdot \Delta\theta + \theta_{\max} + \text{rand}(\theta_{\text{rand}}) \quad \text{(A.5)}$$

with $\Delta\theta$ the difference in tilt between the smallest and largest leaves (fixed at 20°). $\theta_{\max}$ is the leaf inclination angle of the largest leaf (fixed at 45°). The term rand(θ_{rand}) ($-\theta_{\text{rand}}/2 < \text{rand}(\theta_{\text{rand}}) < \theta_{\text{rand}}/2$) introduces some randomness. The normalized difference between the largest and smallest leaf (S''_N) is computed as

$$S''_N = \frac{S'_{N_{\max}} - S'_N}{S'_{N_{\max}}}. \quad \text{(A.6)}$$

Leaf azimuth angles (φ_N) are normally distributed, with leaves growing alternately on both sides, requiring a Gaussian standard deviation (φ_{rand}) of the azimuth angle.

The attachment point heights of several leaf order N are [27]

$$H_N = \frac{H_{\max} - H_1}{(N_{\max})^k - 1} \cdot \left(N^k - 1\right) + H_1 \quad \text{(A.7)}$$

with H_1, the height of the first leaf uses a fixed value of 0.015 m. $H_{\max}$ is the maximum plant height defined by the user, and k is calculated as follows:

$$k = 3.7 - 0.1 \cdot N_{\max} - 0.36 \cdot S_{\max}. \quad \text{(A.8)}$$

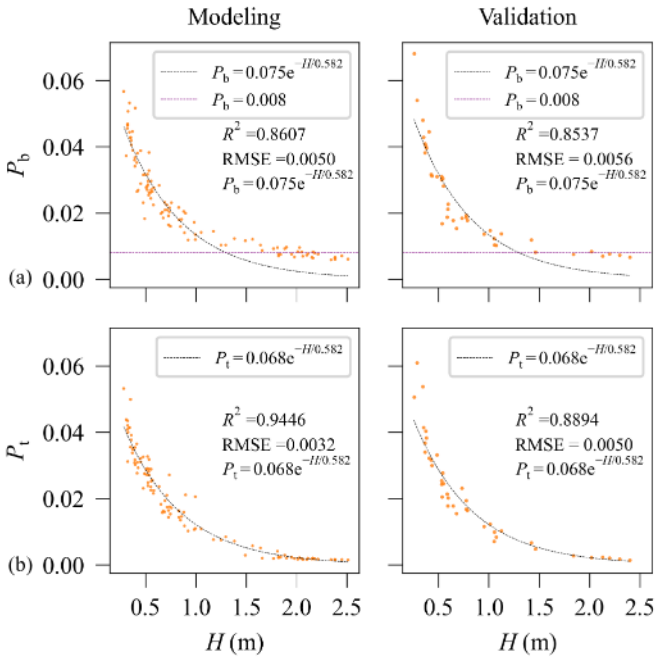


Fig. 19. Exponential relationship between (a) stem ratio (P_b) of bottom radius (R_b) and height (H) and between (b) stem ratio (P_t) of top radius (R_t) and H for waterlogging experiment data. Modeling (left) and validation (right) groups with data randomly assigned to these groups in a 7:3 ratio.

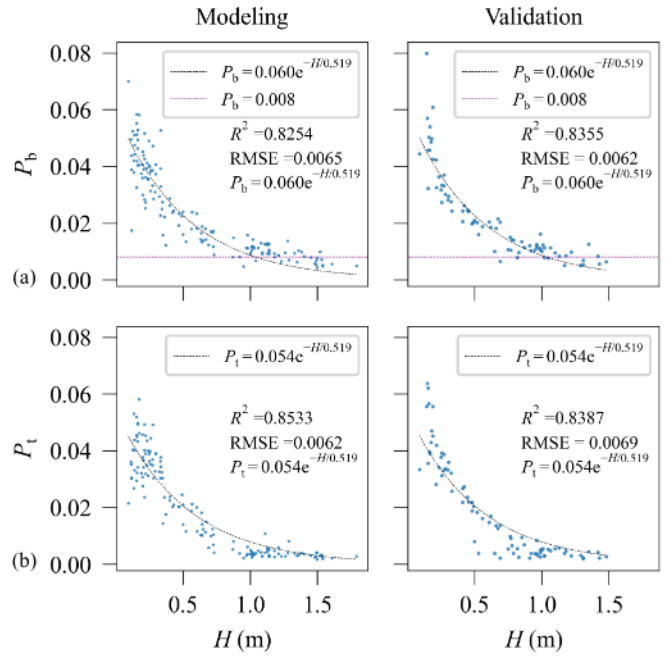


Fig. 20. Exponential relationship between (a) stem ratio (P_b) of bottom radius (R_b) and height (H) and between (b) stem ratio (P_t) of top radius (R_t) and H for fertilizer experiment data. Modeling (left) and validation (right) groups with data randomly assigned to these groups in a 7:3 ratio.

The radius of the stem (R_N) at where the leaf order N is attached is calculated as a linear function

$$R_N = R_1 + \left(\frac{R_{\max} - R_1}{N_{\max} - 1}\right) \cdot (N - 1) \quad \text{(A.9)}$$

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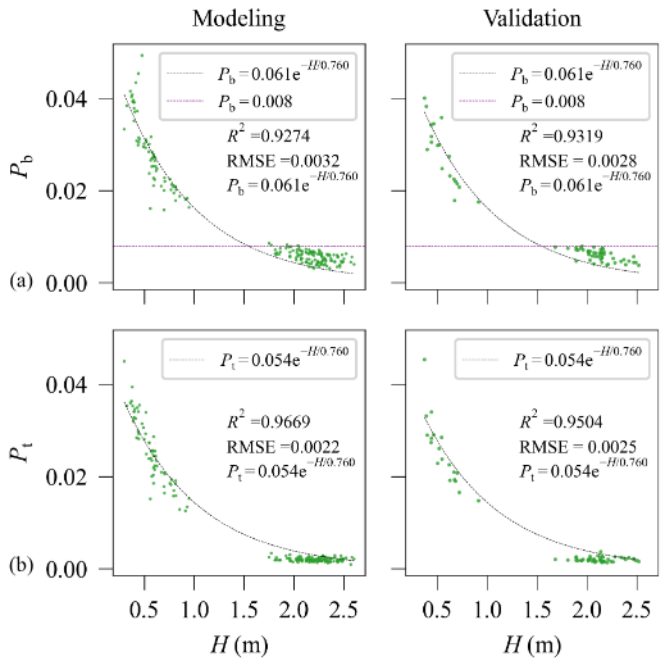


Fig. 21. Exponential relationship between (a) stem ratio (P_b) of bottom radius (R_b) and height (H) and between (b) stem ratio (P_t) of top radius (R_t) and H for drought experiment data. Modeling (left) and validation (right) groups with data randomly assigned to these groups in a 7:3 ratio.

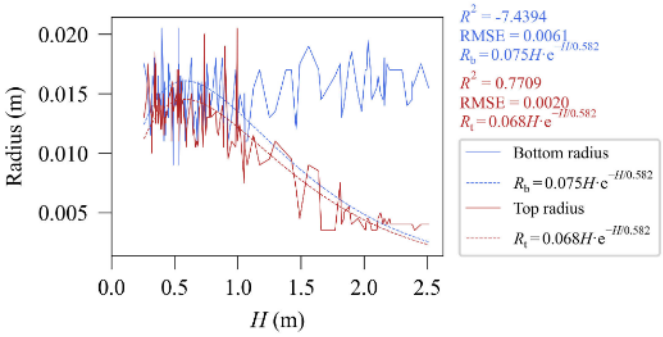


Fig. 22. Relationship between stem radius and height (H) of waterlogging experiment data. Blue is for the bottom stem radius (R_b), and red is for the top stem radius (R_t). Solid lines represent the measured data, and dashed lines represent the trend line of the fitting equations.

with R_1, the radius of the stem in the first phase of the leaf uses a fixed value of 0.002 m. The maximum radius of the stem ($R_{\max}$) is proportional to $H_{\max}$ and is calculated as

$$R_{\max} = 0.008 \cdot H_{\max}. \tag{A.10}$$

The number of leaves existing in the plant was determined according to the cumulative temperature required to grow the first leaf, the cumulative temperature required to grow the next leaf, and the cumulative temperature time the leaf can survive. We calculated the relevant geometric parameters according to (A.1)–(A.10). These geometric parameters were then used in the ELSYS as ELSYS coupling 4-D maize ($\text{ELSYS}_{4\text{Dmaize}}$) model.

Appendix B

We separately calculated the fitting error metrics for each stress experiment (waterlogging, fertilizer, and drought) from

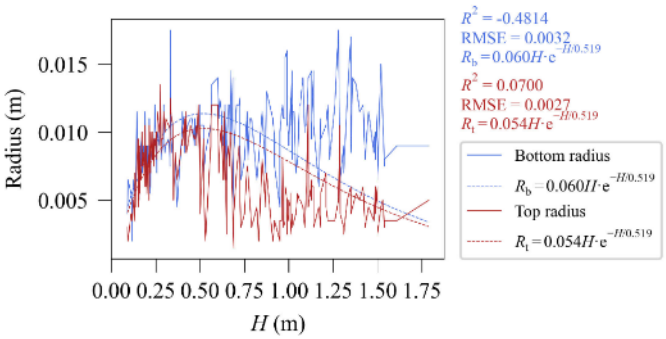


Fig. 23. Relationship between stem radius and height (H) of fertilizer experiment data. Blue is for the bottom stem radius (R_b), and red is for the top stem radius (R_t). Solid lines represent the measured data, and dashed lines represent the trend line of the fitting equations.

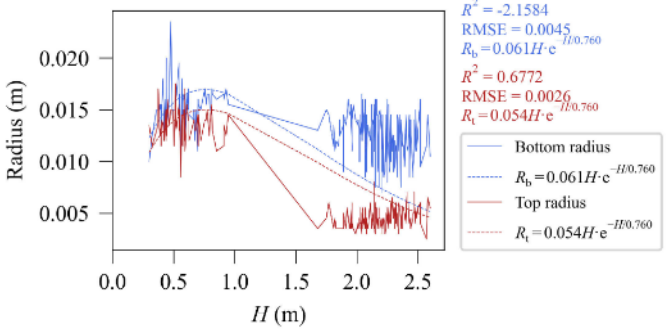


Fig. 24. Relationship between stem radius and height (H) of drought experiment data. Blue is for the bottom stem radius (R_b), and red is for the top stem radius (R_t). Solid lines represent the measured data, and dashed lines represent the trend line of the fitting equations.

the stress-impacted group [Figs. 5 and 6(b)]. The exponential relationship between stem ratio (P_b for bottom radius R_b and P_t for top radius R_t) and height H for all three cases (Figs. 19–21) exhibited strong fits. However, R^2 for the relationship between radius and height (H) remained relatively low (Figs. 22–24). The underlying reason for this lower R^2 might be attributed to the influence of stressors that impeded the growth of maize stem radius. The precise nature and extent of these impacts were notably intricate. For instance, experiments involving pressure on the top stem radius appeared to exert a more pronounced growth inhibition than those on the bottom stem (Figs. 22–24). Furthermore, it was apparent that the location of the inflection points within the function lacked uniformity across various situations. In our dataset, the waterlogging experiment (Fig. 22) reached the inflection point earliest, followed by the fertilizer experiment (Fig. 23), whereas the drought experiment (Fig. 24) appeared to attain the inflection point at a later stage.

Given the diversity in experimental conditions designed for this study, devising a generalized function to describe these observed changes comprehensively remains challenging. Our study aimed to develop a model for normal growth maize; therefore, we were predisposed to construe this facet of the experimental findings as a constraint inherent to the model, emphasizing the need for adjustments to the growth equations to accommodate nonnormal growth scenarios for the DLAmaize model. However, as identified in this study,

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the exponential relationship between stem radius–height ratio (P_b and P_t) and height yielded satisfactory fitting results even for the stress-impacted group.

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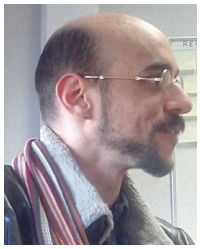
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